

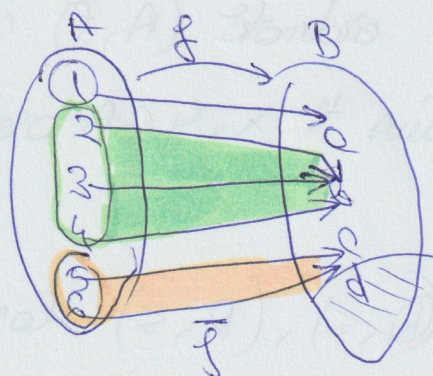
Theorem I de factorizare

! Ex - examen !

$$f: A \rightarrow B \text{ unde } A = \{1, 2, \dots, 6\}$$

$$B = \{a, b, c, d\}$$

x	1	2	3	4	5	6
$f(x)$	a	b	b	b	c	c



$$\bar{f}: A \rightarrow \text{Im}f = \{a, b, c\} \subseteq B$$

$$\bar{f}(x) = f(x) \quad \forall x \in A \text{ unde } \bar{f} \text{ surj}$$

$$\text{Ker}f = (A, A, \text{Ker}f), \quad \text{Ker}f = \{(1,1), (2,2), (2,3), (2,4)$$

$$(3,2), (3,3), (3,4), (4,2),$$

$$(4,3), (4,4), (5,5), (5,6),$$

$$(6,5), (6,6)\}$$

$$[1] \stackrel{\text{def}}{=} \text{Ker}f < 1 > = \{1\}$$

$$[2] \stackrel{\text{def}}{=} \text{Ker}f < 2 > = \{2, 3, 4\} = [3] = [4]$$

$$[5] \stackrel{\text{def}}{=} \text{Ker}f < 5 > = \{5, 6\} = [6]$$

$$A/\text{Ker}f = \{[1], [2], [5]\} = \{\{1\}, \{2, 3, 4\}, \{5, 6\}\}$$

$$\tilde{f}: A/\text{Ker}f \rightarrow \text{Im}f$$

$[x]$	$\{1\}$	$\{2, 3, 4\}$	$\{5, 6\}$
$\tilde{f}([x])$	a	b	c

 $\Rightarrow f \text{ bijectiv}$

$$f([x]) = f(x) \quad \forall x \in A$$

Multimi ordonate

Recapitulare: Def: O relatie omogenă $\rho = (A, A, R)$ s.m. rel de ordine dacă ρ este $R + T + AS$.

În această situație spunem că (A, ρ) este o multime ordonată

Def: O multime ordonată (A, ρ) s.m. total ordonată sau lont dacă în plus $\forall x, y \in A$ avem că $x \rho y$ sau $y \rho x$.

Ex: • $(\mathbb{N}, \leq)$, $(\mathbb{Z}, \leq)$, $(\mathbb{Q}, \leq)$, $(\mathbb{R}, \leq)$ lonturi

• $(\mathbb{N}, |)$ mult ordonată dar NU este lont: (ex: $2 \nmid 3$ și $3 \nmid 2$)

• $(\mathbb{Z}, |)$ nu este nici multime ordonată pt „ $|$ ” pe $\mathbb{Z}$
NU este rel de ordine (NU avem AS: $2 \mid -2$ și $-2 \nmid 2$ dar $-2 \nmid 2$)

• Fie A o mult $(P(A), \subseteq)$ este mult ord pt $|A| \geq 2$ NU este lont,

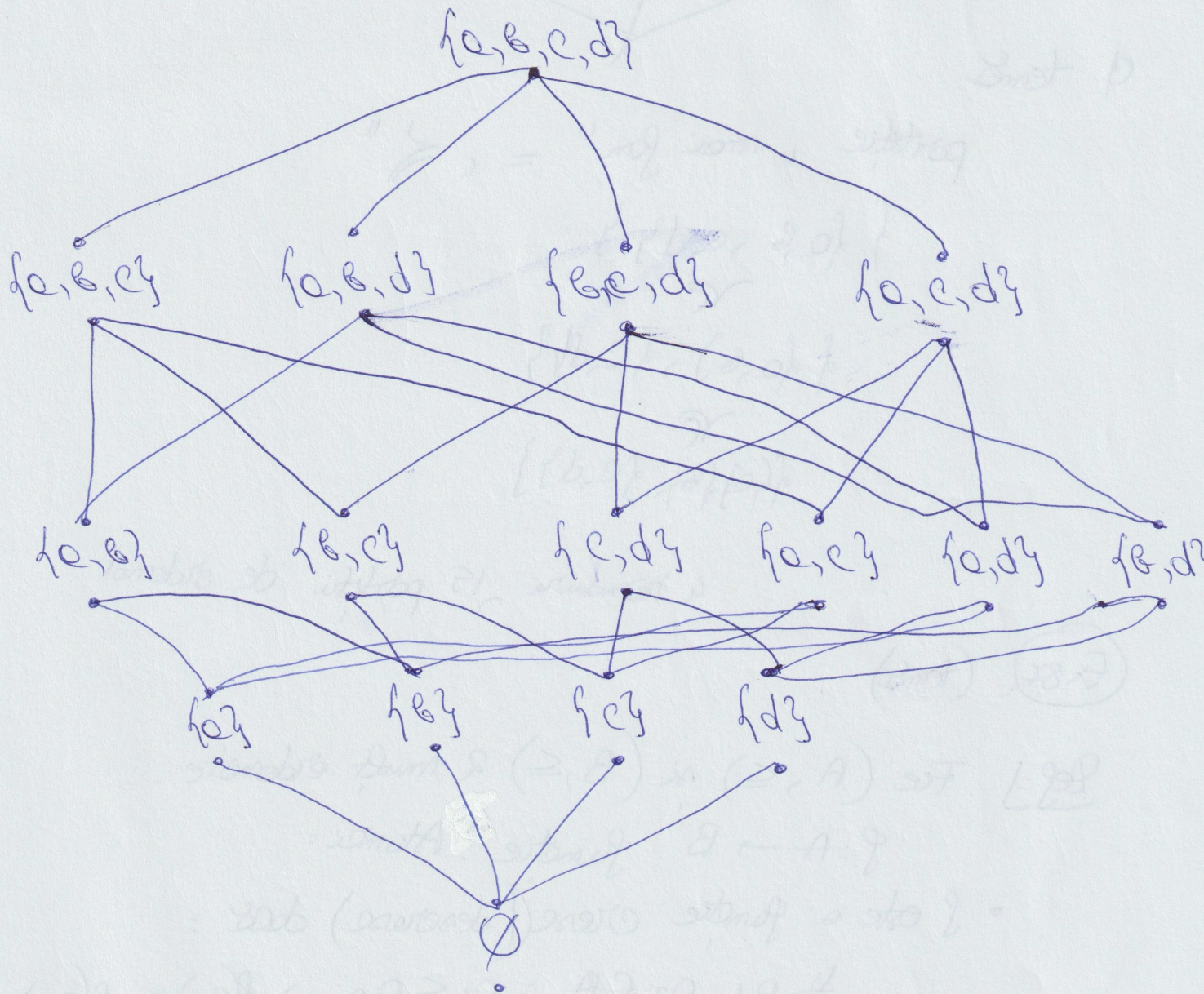
$$\text{Not} = \mathcal{O}(A) = \{ \rho = (A, A, R) \mid \rho \text{ rel de ordine} \}$$

Ex 88 Se se întocmească diagramele Hasse ale urm. multimi ordonate:

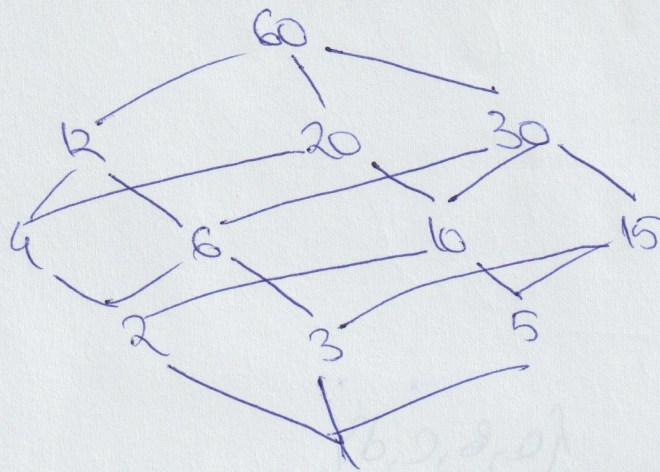
a) $(P(\{a, b, c, d\}), \subseteq)$

Rechner

a)



b) $(D_{60}, 1)$ $D_{60} = \{1, 60, 2, 30, 3, 20, 4, 15, 5, 12, 6, 10\}$
 $60 = 2^2 \cdot 3 \cdot 5 \Rightarrow \text{"etwas"}$



9 termi

partitie "mai fan" = " $\preceq$ "

$\{ \{a, b\}, \{c, d\} \}$
 $\preceq$

$\{ \{a, b\}, \{c, d\} \}$

$\preceq$
 $\{ \{a, b\}, \{c, d\} \}$

4 ramduri, 15 partitii de ordinat

Ex 81 (termi)

Def: Fie $(A, \leq)$ si $(B, \leq)$ 2 mult. ordinate.

$f: A \rightarrow B$ functie. Atunci:

- f este o functie cresc (descrasc) daca:

$$\forall a_1, a_2 \in A : a_1 \leq a_2 \rightarrow f(a_1) \leq f(a_2)$$

(respectiv $f(a_2) \leq f(a_1)$)

- este un izomorfism de ordine (oxomomorfism) daca:

1) f cresc. + f bij + f^{-1} cresc

Ex 91) Sei $(A, \leq), (B, \leq)$ mult. ord. u. sei $f: A \rightarrow B$ eine Funktion bij. u. cresc.

a) ~~Sei~~ A total ord. u. f^{-1} cresc. u. B total ord. $\Rightarrow$ 2 sunt comparabile u. dem.

pp A total ordonat u. dem u. $f^{-1}: B \rightarrow A$ este cresc.

Sei $b_1, b_2 \in B$ arbitrare u. r. $b_1 \leq b_2$. dem u.

$$f^{-1}(b_1) \leq f^{-1}(b_2)$$

$$f^{-1}(b_1) = a_1 \in A \Rightarrow f(a_1) = b_1$$

$$f^{-1}(b_2) = a_2 \in A \quad f(a_2) = b_2$$

$$\left. \begin{array}{l} (A, \leq) - \text{total ordonat} \\ a_1, a_2 \in A \end{array} \right\} \Rightarrow a_1 \leq a_2 \text{ sau } a_2 \leq a_1$$

$$\underline{I}: \left. \begin{array}{l} a_1 \leq a_2 \\ f \text{ cresc} \end{array} \right\} \Leftrightarrow f^{-1}(b_1) \leq f^{-1}(b_2) \quad \checkmark$$

$$\underline{II}: \left. \begin{array}{l} a_2 \leq a_1 \\ f \text{ cresc} \end{array} \right\} \Rightarrow f(a_2) \leq f(a_1) \Rightarrow b_2 \leq b_1 \quad \left. \begin{array}{l} b_2 \leq b_1 \\ b_1 \leq b_2 \end{array} \right\} \Rightarrow b_1 = b_2$$

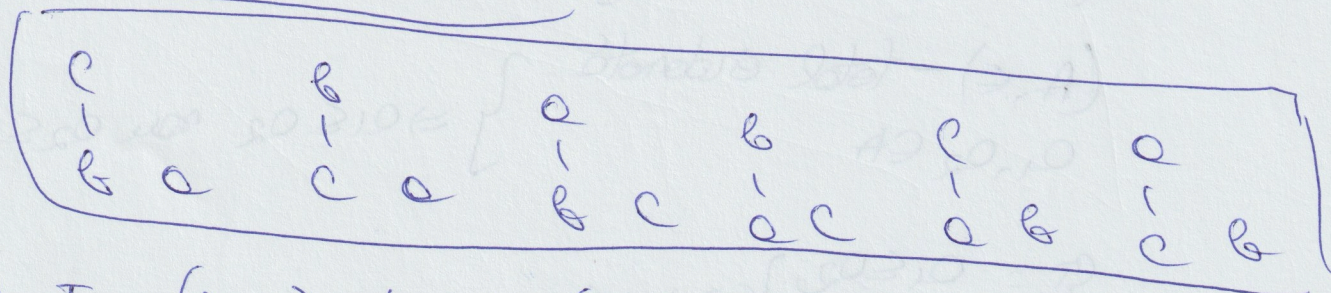
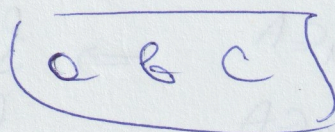
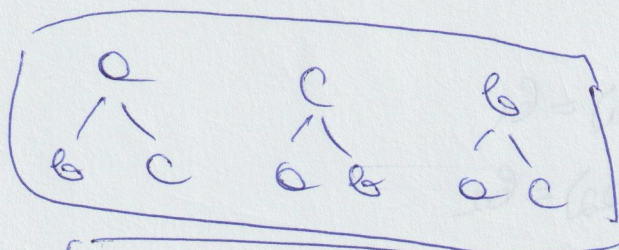
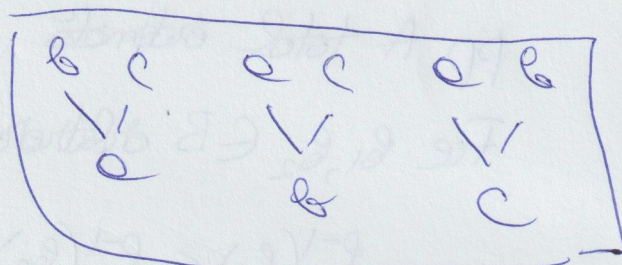
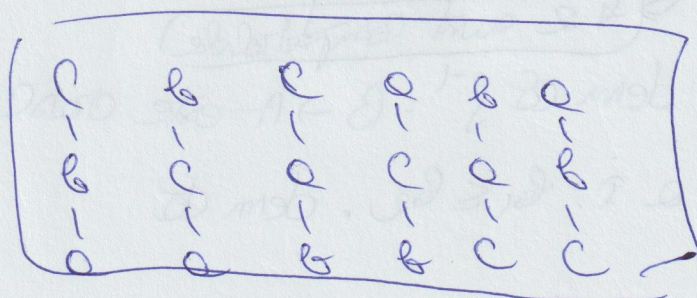
$$\Rightarrow b_1 = b_2 \Rightarrow a_1 = a_2 \Rightarrow a_1 \leq a_2 \Leftrightarrow f^{-1}(b_1) = f^{-1}(b_2)$$

b) B total ord.

2 elem. or. u. sunt comparabile

teme + (94)

Ex 89) Se se det toate rel de ordine definite pe $A = \{a, b, c\}$ (folosind diagrame Hasse). Se se imparte acestea in clase de echivalenta



Ex 90) Fie $(A, \leq)$, $(B, \leq)$, $(C, \leq)$ multimi ordonate si
 fie $f: A \rightarrow B$, $g: B \rightarrow C$ functii.
 a) Daca f si g cresc (si descreasc) at $g \circ f$ cresc
 pp f si g ~~creasc~~ - dem $g \circ f = A \rightarrow C$ cresc

Fie $a_1, a_2 \in A$ oarecare at $a_1 \leq a_2$

Dem ca $(g \circ f)(a_1) \leq (g \circ f)(a_2)$

Cum $a_1 \leq a_2 \overset{\text{in } A}{\mid} f \Rightarrow f(a_1) \leq f(a_2) \overset{\text{in } B}{\mid} g \Rightarrow g(f(a_1)) \leq g(f(a_2))$

$\Rightarrow (g \circ f)(a_1) \leq (g \circ f)(a_2)$